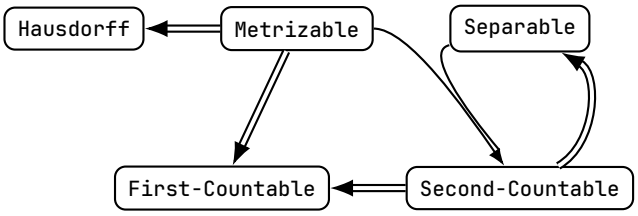
### 0.0.1 Metrizable Theorem Diagram




| **Topology for $\mathbb{R}$:** $\mathcal{T}_{id} \subset \mathcal{T}_f \subset \mathcal{T}_u \subset \mathcal{T}_l \subset \mathcal{T}_d,\ \mathcal{T}_f \subset \mathcal{T}_c$ | | | | | | |
|---|---|---|---|---|---|---|
| Topology | $\mathcal{T}_{id}$ | $\mathcal{T}_f$ | $\mathcal{T}_u$ | $\mathcal{T}_l$ | $\mathcal{T}_d$ | $\mathcal{T}_c$ |
| First-Countable | Yes | No | Yes | Yes | Yes | No |
| Second-Countable | Yes | No | Yes | No | No | No |
| Hausdorff | No | No | Yes | Yes | Yes | No |
| Metrizable | No | No | Yes | No | Yes | No |
| Separable | Yes | Yes | Yes | Yes | No | No |
| Compactness | Yes | Yes | No | No | No | No |

Let a Topological Space $(X, \mathcal{T})$ be given. For $X$, the following theorems hold:

**Theorem 0.0.1.** If Metrizable, then Hausdorff.

*Proof.* Let $(X, \mathcal{T})$ be a metrizable topological space. Then, there is a metric $d : X \times X \to \mathbb{R}$.
Let distinct two point $x, y \in X$ be given. And, put $\varepsilon = \frac{d(x,y)}{2} > 0$.
Now, $x \in B_d(x; \varepsilon)$ and $y \in B_d(y; \varepsilon)$ but there intersection is empty set because:
Choose $p \in B_d(x; \varepsilon) \cap B_d(y; \varepsilon)$. Then, $d(x, p), d(y, p) < \varepsilon$.
But, this implies that

$$d(x, y) \leq d(x, p) + d(p, y) < 2\varepsilon = d(x, y)$$

Thus, contradiction. Consequently, given topological space satisfies Hausdorff property. □

**Lemma 0.0.2.** Let $(X, d)$ be a Metric Space.
For any $U \in \mathcal{T}_d$, $x \in U$, there is $\delta > 0$ such that $x \in B_d(x, \delta) \subset U$.

*Proof.* Let $U \in \mathcal{T}_d$ be given.
Then, there exist $y \in X$ and $\varepsilon > 0$ such that $x \in B_d(y, \varepsilon) \subset U$. For $\delta = \varepsilon - d(x, y) > 0$, $B_d(x, \delta) \subset B_d(y, \varepsilon)$ because:

$$\begin{aligned} p \in B_d(x, \delta) &\iff d(p, x) < \delta \\ &\implies d(p, y) \leq d(p, x) + d(x, y) < \delta + d(x, y) = \varepsilon \\ &\iff p \in B_d(y, \varepsilon) \end{aligned}$$

□

**Theorem 0.0.3.** If Metrizable, then First-Countable.

*Proof.* Let $(X, \mathcal{T})$ be a metrizable topological space, and $d$ be a metric that induces $\mathcal{T}$.
For any $x \in X$, define

$$\beta_x \overset{\text{def}}{=} \left\{ B_{\frac{1}{n}}(x) \;\middle|\; n \in \mathbb{N} \right\}$$

Trivially, $\beta_x$ is countable set. Want To Show: $\beta_x$ be a local base of $x$.
**LB1**) is clear.
**LB2**) Let $U \in \mathcal{T}$ with $x \in U$ be given. Since above lemma, there is $\delta > 0$ such that $x \in B_d(x, \delta) \subset U$.
Let $n \in \mathbb{N}$ be a number such that $1 < \delta n$, then $B_d\left(x, \frac{1}{n}\right) \in \beta_x$ satisfies

$$x \in B_d\left(x, \frac{1}{n}\right) \subset B_d(x, \delta) \subset U$$

□

**Theorem 0.0.4.** If Metrizable and Separable, then Second-Countable.

*Proof.* Since $X$ is Separable, let $A = \{x_i \mid i \in \mathbb{N}\}$ be a countable dense subset of $X$.
Let $d : X \times X \to \mathbb{R}$ be a metric on $X$ that induces $\mathcal{T}$. Define

$$\beta \stackrel{\text{def}}{=} \left\{ B_d\left(x_i, \frac{1}{n}\right) \;\middle|\; i, n \in \mathbb{N} \right\}$$

Clearly, $\beta$ is Countable and $\beta \subset \mathcal{T}$.
Enough to Show that: For any $U \in \mathcal{T}$, there exists $\mathcal{C} \subset \beta$ such that

$$U = \bigcup_{u \in \mathcal{C}} u$$

Let $U \in \mathcal{T}$ be given. Since $\mathcal{T}$ is consisted by open balls induced a metric $d$,
for any $x \in U$, there exists $r > 0$ such that $B_d(x, r) \subset U$ by the lemma.
Set $\delta_x = \frac{1}{2n}$ such that $1 < nr$. Since $A$ is dense in $X$,

$$A \cap B_d\left(x, \delta_x\right) \neq \emptyset$$

That is, for some $i \in \mathbb{N}$, $d(x, x_i) < \delta_x$. Now,

$$x \in B_d\left(x_i, \delta_x\right) \subset B_d(x, 2\delta_x) \subset B_d(x, r) \subset U$$

Consequently, we shown that:

$$\forall x \in U,\ \exists B_x \in \beta \ \text{ s.t } \ x \in B_x \subset U$$

Now, Construct given $U \in \mathcal{T}$ as: Since **Axiom of Choice**, Define $\mathcal{C} \stackrel{\text{def}}{=} \bigcup_{x \in U} \{B_x \in \beta \mid x \in B_x \subset U\}$. Then,

$$U = \bigcup_{u \in \mathcal{C}} u$$

□

**Theorem 0.0.5.** If Second-Countable, then Separable.

*Proof.* Let $\beta = \{B_i \mid i \in \mathbb{N},\ B_i \neq \emptyset\}$ be a Countable Basis of $X$.
By the **Axiom of Choice**, we can construct a countable set $A = \{x_i \mid i \in \mathbb{N},\ x_i \in B_i\}$. WTS: $\bar{A} \supset X$.
Let $x \in X$ and $U \in \mathcal{T}$ with $x \in U$ be given arbitrarily.
Since $\beta$ is basis of $\mathcal{T}$, $x \in B_i \subset U$ for some $B_i \in \beta$. Thus, $x_i \in U$, so we obtain $A \cap U \neq \emptyset$. Therefore, $x \in \bar{A}$. □

**Theorem 0.0.6.** If Second-Countable, then First-Countable.

*Proof.* Trivial. □